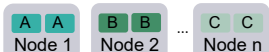


Heterogeneous Cluster

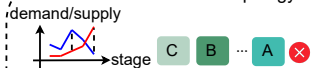


MLLM Configs



MMoH Parallel Strategy Planner

Demand-Driven DeviceTopology



— MLLM comp./mem. demand tendency

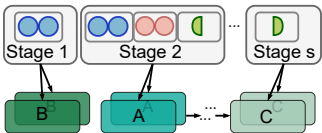
— Device comp./mem. supply tendency

Hybrid-Granularity Abstraction

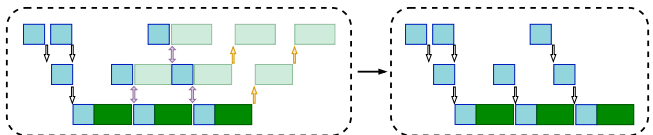


Model Chunk List

Frozen-Aware Model Partitioning



Early-Termination Scheduler



Forward Pass Backward Pass Omissible Computation

↕↕↕ P2P Communication ↕ Alterable Communication ⬆ Omissible Communication